

Digital Skill Premia, Structural Polarization, and Skill Complementarity

Evidence from Chinese Listed-Firm Job Postings, 2016–2025

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2026.4.20



- 1 Introduction & Motivation
- 2 Data & Empirical Strategy
- 3 Time-Varying Wage Premia
- 4 Skill Complementarity
- 5 Structural Polarization & Wage Inequality
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- **Central question:** Does digital technology raise wages, and for whom?
- Classic SBTC framework (Katz & Murphy 1992; Acemoglu 2002): technology raises demand for abstract/complementary skills
- Task-based models (Autor et al. 2003; Acemoglu & Autor 2011): technology may also *polarize* — substituting routine middle-skill tasks

Why China?

- 1 Exceptionally rapid tech adoption 2016–2025 (cloud → big data → LLMs)
- 2 Sharp distributional tensions: massive low-skill supply + booming college workforce
- 3 Listed-firm job postings: high-frequency employer-side wage signal



1 Time-varying premium dynamics

Annual regressions reveal a **hump-shaped** AI/ML premium trajectory (35% $\rightarrow$ 55% $\rightarrow$ 25%), interpreted as demand shock followed by lagged supply response

2 Skill complementarity

All pairwise skill interactions are **negative** (sub-additive): holding two digital skills pays less than the sum of individual premia

3 Structural polarization & wage inequality

Industry digital-skill demand *diffused* (Gini: 0.20 $\rightarrow$ 0.14), yet upper-tail wage inequality widened; quantile regression unpacks why



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Sample

- 6.93 million job postings from Chinese listed firms, 2016–2025
- *Wage analysis sample*: 820,807 obs. with valid salary
- 57,460 distinct firms; 3,824 city×year cells
- Salary standardized to monthly midpoint, log form

Skill Classification

- Structured 13-category NLP dictionary
- Core digital skills:
 - ▶ AI / Machine Learning (1.3%)
 - ▶ Programming (10.6%)
 - ▶ Data Engineering (4.8%)
 - ▶ Cloud / DevOps (3.5%)
 - ▶ Data Analysis (21.6%)
- *HasDigital* composite: 79.8%



OLS Wage Regression

$$\ln w_{icft} = \alpha + \sum_k \beta_k S_{icft}^k + \gamma_1 Edu_{it} + \gamma_2 Exp_{it} + \lambda_{ct} + \mu_j + \varepsilon_{icft}$$

- λ_{ct} : **city×year fixed effects** (3,824 groups) absorb local labor-market shocks
- μ_j : **industry FE** (21 categories)
- Standard errors **clustered at firm level** ($G = 57,460$)

Estimation: City×year FE absorbed via iterative within-transformation (Mundlak–Frisch–Waugh, 4 rounds) — equivalent to full dummy inclusion but $O(N)$ per round



Annual regressions

Estimate $\hat{\beta}_k(t)$ year by year to trace premium dynamics

Interaction model

$$+ \sum_{k < k'} \theta_{kk'} S^k S^{k'}$$

$\theta_{kk'} < 0$: sub-additivity

$\theta_{kk'} > 0$: super-additivity

Quantile regression

Estimate at $\tau \in \{0.10, 0.25, 0.50, 0.75, 0.90\}$ to characterize distributional effects

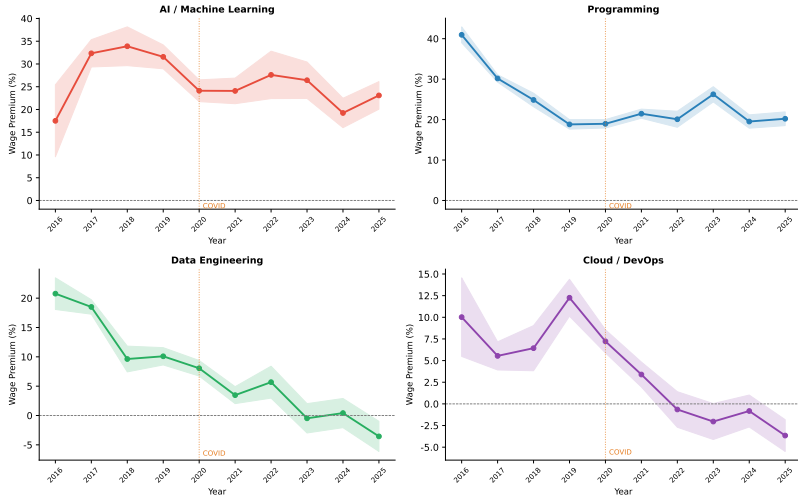


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Annual Wage Premia, 2016–2025



Figure 5: Time-Varying Digital Skill Wage Premia (2016–2025)
Annual OLS with 95% CI; Controls: Education, Experience, City Tier, Industry FE





■ AI/ML — hump-shaped:

- ▶ $\approx 35\%$ in 2016 $\rightarrow$ peak $\approx 55\%$ in 2018–2019 $\rightarrow \approx 25\%$ by 2022–2023
- ▶ Lagged supply response: AI program graduates entered market ≈ 2020
- ▶ Post-2023 partial recovery: generative AI demand wave

■ Programming — gradual moderation: 30% $\rightarrow$ 20% (steady supply growth)

■ Data Engineering — U-shaped: Hadoop era (2016–17) $\rightarrow$ trough $\rightarrow$ stream-processing era (Flink/Kafka post-2021) recovery

Policy Implication

Cross-sectional studies sampled near a peak vs. trough yield very different conclusions — temporal dynamics matter



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Pairwise Interaction	$\hat{\theta}_{kk'}$	SE
AI $\times$ Programming	-0.048***	(0.014)
AI $\times$ Data Eng.	-0.106***	(0.016)
Programming $\times$ Data Eng.	-0.229***	(0.009)
Programming $\times$ Cloud	-0.095***	(0.009)
AI $\times$ Cloud	-0.097***	(0.021)

Implied joint premia:

- AI + Programming: additive 61.1% $\rightarrow$ realized 57.1%
- AI + Data Eng.: additive 62.3% $\rightarrow$ realized 46.1%
- Prog. + Data Eng.: additive 55.5% $\rightarrow$ realized 32.7%

Mechanisms:

- 1 *Task substitutability*: overlapping ETL/query tasks
- 2 *Employer signaling skepticism*: multi-skill claims discounted



O-ring theory prediction (Kremer 1993)

Super-additivity: simultaneous competence in multiple domains should multiply output

Our finding: Sub-additivity

All five pairwise interactions are **negative and significant**

Largest gap: Programming $\times$ Data Engineering (-22.9 pp)

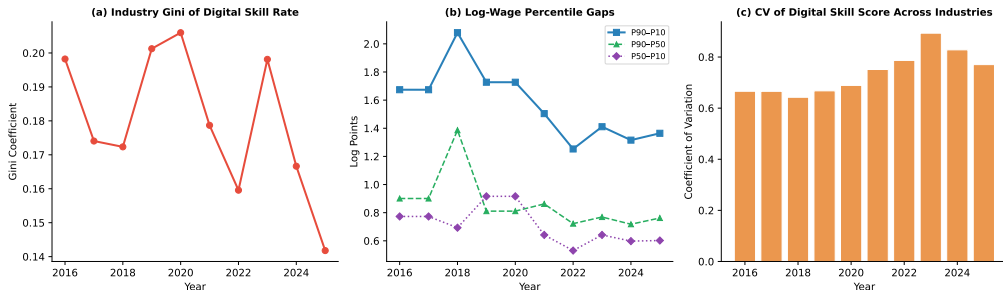
Curriculum design implication:

- Mastering *one* high-premium skill (e.g., AI/ML) dominates spreading effort across multiple domains
- Optimal skill portfolio: concentrated, not diversified



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Figure 6: Polarization of Digital Skill Demand and Wage Inequality (2016-2025)



(a) Industry Gini of digital-skill demand rate. (b) P90–P10/P90–P50/P50–P10 log-wage gaps. (c) CV of digital-skill score across industries.



Demand side: diffusion

- Industry Gini fell from 0.20 (2016) to 0.14 (2025) — -30%
- Digital skills spread from tech-intensive sectors to the broader economy

Expected implication:

Broader demand $\Rightarrow$ compressed inequality

Wage side: upper-tail widening

- P90–P10 log-wage gap: 1.67 (2016) $\rightarrow$ peak 2.08 (2018) $\rightarrow$ >1.25 (2025)
- P90–P50 gap dominates; P50–P10 gap stable
- Upper-tail polarization, not symmetric spreading

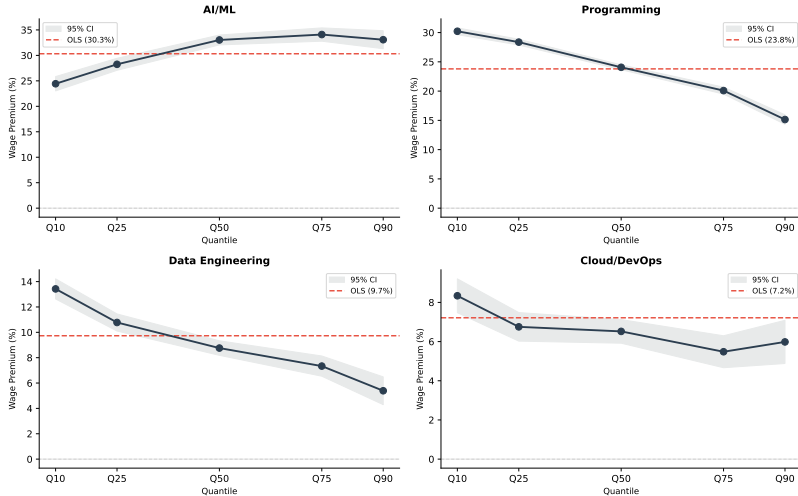
Paradox: diffusion $\neq$ compressed inequality

Diffusion of demand does not guarantee compression of wages when returns are heterogeneous across the distribution

Quantile Regression: Resolving the Paradox



Figure 9: Quantile Regression — Skill Wage Premia Across the Wage Distribution
(Shaded area = 95% CI; dashed line = OLS benchmark)





AI / ML

- Q10: 25.3% → Q75: 34.8% → Q90: 33.1%
- Inverted-U: premium concentrates in high-tier technical roles
- OLS $\approx$ median estimate

Programming

- Q10: 31.2% → Q90: 16.1% (monotone decline)
- Diffused into mid-to-lower-wage positions
- Scarcity rent at the bottom: rare skill in low-skill context

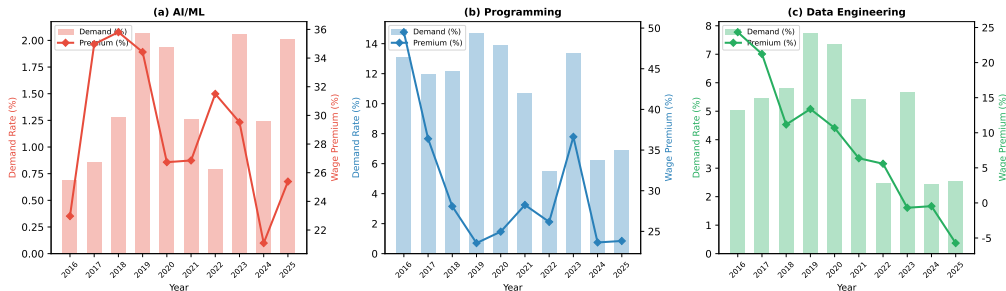
Explanation

Extensive-margin diffusion of skills co-exists with intensive-margin amplification at specific wage quantiles — both mechanisms operate simultaneously

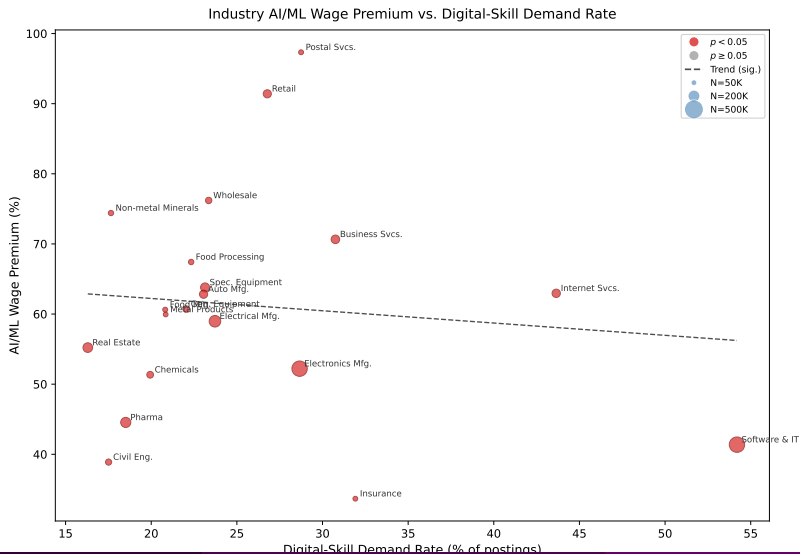
Demand–Premium Decomposition



Figure 8: Decomposition of Skill Demand Rate vs. Wage Premium (2016-2025)



Bars (left): annual demand rate. Line (right): conditional wage premium. AI/ML: demand grows monotonically while premium peaks then retreats — classic lagged supply response.





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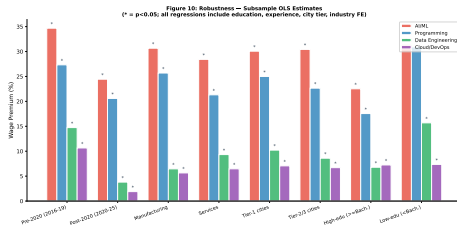
Subsample	N	AI/ML	Programming	Data Eng.
Full baseline	820,807	26.0%***	24.8%***	9.9%***
Pre-2020 (2016–19)	433,018	29.7%***	27.6%***	13.6%***
Post-2020 (2020–25)	387,789	23.7%***	20.4%***	4.2%***
Manufacturing	378,413	26.7%***	27.1%***	8.9%***
Services	285,803	24.9%***	21.8%***	8.9%***
Tier-1 cities	285,151	26.3%***	25.7%***	10.5%***
Tier-2/3 cities	535,656	25.6%***	23.2%***	8.6%***
High edu ($\geq$ BA)	397,319	19.2%***	18.0%***	6.8%***
Low edu ($<$ BA)	423,488	25.7%***	33.3%***	16.6%***

- All premia positive and significant in every subsample
- Data engineering and cloud compress sharply post-2020; AI and programming persist
- Programming premium *higher* for low-education workers: scarcity rents

Robustness: Salary Trim Sensitivity

Salary Trim	N	AI/ML	Programming	Data Eng.
No trim [0%, 100%]	820,807	26.0%***	24.8%***	9.9%***
[1%, 99%]	804,975	25.8%***	23.8%***	9.9%***
[5%, 95%]	739,391	20.2%***	23.4%***	9.7%***
[10%, 90%]	671,137	14.9%***	21.9%***	9.0%***

- AI premia decline more steeply under trimming: partially driven by high-paying outliers
- Programming and data engineering premia more stable
- OLS lies close to median quantile estimate for AI/ML





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1 Time-varying premia

AI/ML premium is hump-shaped (2016–2025), consistent with demand shock + lagged supply response. Cross-sectional studies are sensitive to sampling timing.

2 Sub-additive skill combinations

All pairwise digital-skill interactions are negative; largest for Programming $\times$ Data Eng. (−22.9 pp). Invest *deeply* in one high-value skill.

3 Diffusion $\neq$ equality

Industry digital demand diffused (−30% Gini), but upper-tail wage inequality widened. AI/ML premia peak at Q75; programming premia fall from Q10 to Q90.



(i) Curriculum timing

Higher-education systems adapt with 3–5 year lags. Risk: overproducing workers for skills with already-compressing premia

(ii) Depth over breadth

Sub-additivity supports focused mastery over broad digital literacy in skill-formation policy

(iii) Redistribution alongside diffusion

Broader digital-skill demand does not automatically narrow wages. Complementary redistributive instruments are needed

Thank You!

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Comments welcome